Nanyang Technological University
School of Electrical & Electronic Engineering
IE2106 Engineering Mathematics I

Tutorial 5 Probability & Statistics

1. Suppose you purchase 50,000 electrical switches and the supplier guarantees that the shipment will contain no more than 0.1% defectives. To check the shipment, you randomly sample 500 switches, test them, and find that 4 are defective. Assuming the supplier's claim is true, compute the mean and the standard deviation for the number of defectives in a sample of 500 switches. If the supplier's claim is true, is it likely that you have found 4 defective switches in the sample? Based on this sample, what inference would you make concerning the supplier's guarantee?

2. Out of the TV sets sold by an appliance company, 10% require repair before the warranty expires. Using the normal approximation to the binomial distribution, find the probability that the company will have to repair at least 20 of its last 100 sales. What is the probability that the number of TV sets requiring repair is between 6 and 14 inclusive?

3. The weight of a box of cereal has a normal distribution with $\mu = 340$g and $\sigma = 5$g.
   (a) What is the probability that a randomly selected box will weigh between 335g and 345g?
   (b) 10 boxes of cereal are selected at random for weighing. Find the probability that their average weight will be between 335g and 345g.
   (c) 30 boxes of this cereal are selected at random for weighing. Find the probability that the sample variance is more than 36.69.

Answers

1. $\mu = 0.5,\ \sigma = 0.7067$

2. 0.0008, 0.8664

3.(a) 0.6826 (b) 0.9984 (c) 0.05